Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2013-0451-DS-2-RF-102	Issued	2013-12-09

UT Thickness Data Sheet - 2-RF-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 2-RF-102 located within the ATU-100. The objective is to record the re-measurement of the reflux line to ensure wall thickness remains above the minimum allowable limit.

Kevin Doucette, acting as the lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., is responsible for the execution of the measurements and the recording of the data.

Technical Specifications

The subject line, 2-RF-102, is constructed from A312 TP316 with an outer diameter of 60.3 mm and a nominal thickness of 3.91 mm. The design limits for this line are 517 kPa(g) and 104 °C.

The minimum allowable thickness (t-min) for this line is 2.16 mm. All measurements were performed using the stainless-steel velocity setting as required for A312 TP316 materials.

Procedural Basis

Thickness measurements were performed in accordance with the Halward Inspection Services Ltd. procedure HIS-UT-002 Rev 1 and the QSCP Hamilton Inspection Department procedure QSCP-NDT-001 Rev 2. These procedures govern the site NDE requirements for the CUI program, including the use of the ATU-100 equipment.

The inspection methodology adheres to the requirements of ASTM E797 for the ultrasonic examination of materials.

Inspection Summary

The data recorded herein is part of the re-measurement activities detailed in HIS-2013-0451. This specific action was triggered by the need to re-measure the reflux lines using the correct velocity setting.

The results are evaluated against the t-min of 2.16 mm to determine the remaining life and the next inspection interval based on the established corrosion rates and the Class 2 interval for reflux lines.

Thickness readings, measured 2013-12-09

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-09-09	5.39	5.37	0.006	0.014	>50	2023-12-09	
3-RF-101-02	5.49	2.41	2013-09-09	5.34	5.40	0.003	-0.004	>50	2023-12-09	
3-RF-101-03	5.49	2.41	2013-09-09	5.45	5.37	0.005	0.010	>50	2023-12-09	
3-RF-101-04	5.49	2.41	2013-09-09	5.33	5.41	0.003	-0.010	>50	2023-12-09	
3-RF-101-05	5.49	2.41	2013-09-09	5.26	5.32	0.007	0.010	>50	2023-12-09	
3-RF-101-06	5.49	2.41	2013-09-09	5.25	5.17	0.020	0.030	>50	2023-12-09	
3-RF-101-07	5.49	2.41	2013-09-09	5.43	5.48	-0.003	-0.017	>50	2023-12-09	
3-RF-101-09	5.49	2.41	2013-09-09	5.12	5.26	-0.562	-0.562	>50	2023-12-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-09-09	3.71	3.60	0.018	0.032	44.9	2023-12-09	
2-RF-102-02	3.91	2.16	2013-09-09	3.65	3.70	0.010	0.023	>50	2023-12-09	
2-RF-102-03	3.91	2.16	2013-09-09	3.76	3.79	0.008	0.001	>50	2023-12-09	
2-RF-102-04	3.91	2.16	2013-09-09	3.80	3.82	0.003	-0.001	>50	2023-12-09	
2-RF-102-05	3.91	2.16	2013-09-09	3.69	3.67	0.080	0.080	18.8	2023-05-05	
2-RF-102-06	3.91	2.16	2013-09-09	3.71	3.67	0.161	0.161	9.4	2018-08-22	
2-RF-102-07	3.91	2.16	2013-09-09	3.82	3.72	0.014	0.017	>50	2023-12-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-12-09	_____